

# MAT 103: Numerical Analysis I

## Topic 3: Solutions of Scalar Nonlinear Equations

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*"The purpose of computing is insight, not numbers."* — Richard Hamming, Mathematician

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## 1 Introduction

In Topics 1 and 2 we established *why* numerical methods are needed and *how* errors arise. In this topic we put those ideas to work on one of the most common problems in mathematics and engineering: **finding the roots of an equation**.

Consider these situations:

- A chemical engineer needs to find the temperature  $T$  at which the reaction rate  $k(T) = e^{-T/10} - 0.3T$  equals zero — that is, when  $e^{-T/10} = 0.3T$ .
- A structural engineer needs the buckling load  $P$  of a column, satisfying  $\tan(\sqrt{P}) = \sqrt{P}$ .
- A financial analyst needs the interest rate  $r$  satisfying  $(1 + r)^{12} - 1.1 = 0$ .

In every case, the problem reduces to: **find  $x$  such that  $f(x) = 0$** .

For simple functions (linear, quadratic), we have exact formulas. For the more complex, transcendental, or higher-degree cases that arise in practice, we need **numerical root-finding methods**.

By the end of this topic, you should be able to:

- Reformulate a root-finding problem as  $f(x) = 0$  and locate roots graphically.
  - Apply fixed-point iteration and analyse its convergence.
  - Apply the bisection, secant, Regula Falsi, and Newton–Raphson methods.
  - Apply Newton's method to polynomials using Horner's scheme.
  - Understand the principle behind Bairstow's method for polynomial roots.
  - Use MATLAB or MAPLE to find roots computationally.
-

## 2 Finding Roots of $f(x) = 0$

### 2.1 Definition

#### **i** Definition

A **root** (or **zero**) of a function  $f(x)$  is a value  $x = x^*$  such that:

$$f(x^*) = 0$$

**Reformulating any equation as  $f(x) = 0$ :**

Any equation can be written in the form  $f(x) = 0$  by moving everything to the left side.

| Original Equation    | Reformulation            | $f(x)$                      |
|----------------------|--------------------------|-----------------------------|
| $e^x = 3x + 1$       | $e^x - 3x - 1 = 0$       | $f(x) = e^x - 3x - 1$       |
| $x^3 = 2x + 5$       | $x^3 - 2x - 5 = 0$       | $f(x) = x^3 - 2x - 5$       |
| $\sin(x) = x/2$      | $\sin(x) - x/2 = 0$      | $f(x) = \sin(x) - x/2$      |
| $(1 + r)^{12} = 1.1$ | $(1 + r)^{12} - 1.1 = 0$ | $f(r) = (1 + r)^{12} - 1.1$ |

### 2.2 Locating Roots Graphically

Before applying any numerical method, it is good practice to **sketch**  $f(x)$  or plot  $y = f(x)$  and look for where the curve crosses the  $x$ -axis. This gives us:

1. An estimate of how many roots exist.
2. Approximate locations of the roots.
3. A good initial guess for iterative methods.

The **Intermediate Value Theorem (IVT)** gives us a rigorous way to confirm a root exists:

#### **i** Intermediate Value Theorem (IVT)

If  $f$  is **continuous** on  $[a, b]$  and  $f(a)$  and  $f(b)$  have **opposite signs** (i.e.,  $f(a) \cdot f(b) < 0$ ), then there exists at least one root  $x^* \in (a, b)$  such that  $f(x^*) = 0$ .

#### **Solved Example 3.1 — Locating a root**

Show that  $f(x) = x^3 - 4x + 1$  has a root in each of the intervals  $[-3, -2]$ ,  $[0, 1]$ , and  $[1, 2]$ .

*Solution:*

Evaluate  $f$  at the endpoints of each interval:

| Interval   | $f(\text{left end})$         | $f(\text{right end})$    | Sign change? | Conclusion         |
|------------|------------------------------|--------------------------|--------------|--------------------|
| $[-3, -2]$ | $f(-3) = -27+12+1 = -14 < 0$ | $f(-2) = -8+8+1 = 1 > 0$ | Yes          | Root in $(-3, -2)$ |
| $[0, 1]$   | $f(0) = 1 > 0$               | $f(1) = 1-4+1 = -2 < 0$  | Yes          | Root in $(0, 1)$   |
| $[1, 2]$   | $f(1) = -2 < 0$              | $f(2) = 8-8+1 = 1 > 0$   | Yes          | Root in $(1, 2)$   |

Since  $f$  is a polynomial (continuous everywhere), the IVT guarantees a root in each interval. This function has exactly three real roots.

### 3 General Iteration Methods (Fixed-Point Iteration)

#### 3.1 The Idea

The **fixed-point iteration** method (also called *successive substitution*) converts the equation  $f(x) = 0$  into the equivalent form:

$$x = g(x)$$

We call  $x^*$  a **fixed point** of  $g$  if  $g(x^*) = x^*$ . Clearly, if  $x^*$  is a fixed point of  $g$  and  $g$  was obtained from  $f$ , then  $f(x^*) = 0$ .

Starting from an initial guess  $x_0$ , we generate the sequence:

$$x_{n+1} = g(x_n), \quad n = 0, 1, 2, \dots$$

If this sequence converges, it converges to a fixed point  $x^*$  which is a root of  $f(x) = 0$ .

#### 3.2 How to Rearrange $f(x) = 0$ into $x = g(x)$

There are usually **many ways** to rearrange  $f(x) = 0$  as  $x = g(x)$ . Some rearrangements lead to convergence; others lead to divergence.

**Example:**  $f(x) = x^2 - 3x + 2 = 0$  (roots:  $x = 1$  and  $x = 2$ )

| Rearrangement         | $g(x)$                     | Converges to?               |
|-----------------------|----------------------------|-----------------------------|
| $x = \frac{x^2+2}{3}$ | $g_1(x) = \frac{x^2+2}{3}$ | $x = 1$ (from $x_0$ near 1) |

| Rearrangement         | $g(x)$                     | Converges to?               |
|-----------------------|----------------------------|-----------------------------|
| $x = 3 - \frac{2}{x}$ | $g_2(x) = 3 - \frac{2}{x}$ | $x = 2$ (from $x_0$ near 2) |
| $x = x^2 - 2x + 2$    | $g_3(x) = x^2 - 2x + 2$    | $x = 1$ (from $x_0$ near 1) |

The choice of  $g$  is critical. The convergence condition tells us which choices work.

### 3.3 Convergence Condition for Fixed-Point Iteration

#### **i** Convergence Theorem

Let  $g$  be continuous and differentiable on an interval  $[a, b]$  containing the fixed point  $x^*$ . If:

$$|g'(x)| \leq K < 1 \quad \text{for all } x \in [a, b]$$

for some constant  $K < 1$ , then the iteration  $x_{n+1} = g(x_n)$  **converges** to  $x^*$  for any starting value  $x_0 \in [a, b]$ .

**Interpretation:**  $|g'(x^*)|$  is called the **asymptotic error constant**.

- If  $|g'(x^*)| < 1$ : iteration **converges** — the smaller  $|g'|$ , the faster.
- If  $|g'(x^*)| > 1$ : iteration **diverges**.
- If  $|g'(x^*)| = 1$ : the test is inconclusive.

#### Solved Example 3.2 — Checking convergence

For  $f(x) = x^2 - 3x + 2 = 0$ , check whether  $g_1(x) = \frac{x^2+2}{3}$  converges near  $x^* = 1$ .

*Solution:*

$$g_1'(x) = \frac{2x}{3}$$

At  $x^* = 1$ :  $|g_1'(1)| = \frac{2}{3} < 1$

The condition is satisfied, so the iteration **converges** near  $x^* = 1$ .

Now check  $g_3(x) = x^2 - 2x + 2$  near  $x^* = 1$ :

$$g_3'(x) = 2x - 2$$

At  $x^* = 1$ :  $|g_3'(1)| = |2 - 2| = 0 < 1$

Try  $x_0 = 1.5$ :

$$\begin{aligned}
x_1 &= (1.5)^2 - 2(1.5) + 2 = 2.25 - 3 + 2 = 1.25 \\
x_2 &= (1.25)^2 - 2(1.25) + 2 = 1.5625 - 2.5 + 2 = 1.0625 \\
x_3 &= (1.0625)^2 - 2(1.0625) + 2 = 1.1289 - 2.125 + 2 = 1.0039
\end{aligned}$$

It does converge here! The convergence theorem's condition is checked at the root — and at  $x^* = 1$  the derivative is exactly 0, giving very fast convergence.

### 3.4 Algorithm: Fixed-Point Iteration

Given  $x_0$  (initial guess), tolerance  $\varepsilon > 0$ , and maximum iterations  $N$ :

1. For  $n = 0, 1, 2, \dots, N$ :
  - a. Compute  $x_{n+1} = g(x_n)$
  - b. If  $|x_{n+1} - x_n| < \varepsilon$ : **stop** —  $x_{n+1}$  is the root.
2. If the loop completes without stopping: the method has not converged in  $N$  steps.

### 3.5 Solved Example 3.3 — Full fixed-point iteration

Solve  $f(x) = e^{-x} - x = 0$  using fixed-point iteration.

Rearrange:  $x = e^{-x}$ , so  $g(x) = e^{-x}$ .

Check convergence:  $g'(x) = -e^{-x}$ . At the root  $x^* \approx 0.567$ :  $|g'(0.567)| = e^{-0.567} \approx 0.567 < 1$

Start with  $x_0 = 0$ , tolerance  $\varepsilon = 0.0001$ :

| $n$ | $x_n$    | $x_{n+1} = e^{-x_n}$ | $\ x_{n+1} - x_n\ $             |
|-----|----------|----------------------|---------------------------------|
| 0   | 0.000000 | 1.000000             | 1.000000                        |
| 1   | 1.000000 | 0.367879             | 0.632121                        |
| 2   | 0.367879 | 0.692201             | 0.324322                        |
| 3   | 0.692201 | 0.500474             | 0.191727                        |
| 4   | 0.500474 | 0.606531             | 0.106057                        |
| 5   | 0.606531 | 0.545239             | 0.061292                        |
| 6   | 0.545239 | 0.579703             | 0.034464                        |
| 7   | 0.579703 | 0.560115             | 0.019588                        |
| 8   | 0.560115 | 0.571143             | 0.011028                        |
| 9   | 0.571143 | 0.564879             | 0.006264                        |
| 10  | 0.564879 | 0.568429             | 0.003550                        |
| 15  | 0.567560 | 0.566872             | 0.000688                        |
| 20  | 0.567176 | 0.567370             | 0.000194                        |
| 24  | 0.567296 | 0.567189             | 0.000107                        |
| 25  | 0.567189 | 0.567250             | <b>0.000061</b> $< \varepsilon$ |

**Root:**  $x^* \approx 0.5672$  (the true value is 0.56714...)

The method converges but slowly (linear convergence), taking about 25 iterations. This motivates the faster methods we study next.

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## 4 Convergence Criterion and Order of Convergence

### 4.1 Stopping Criteria

We stop an iterative method when one of the following conditions is met:

**Criterion 1 — Successive approximations are close:**

$$|x_{n+1} - x_n| < \varepsilon_1$$

**Criterion 2 — The function value is close to zero:**

$$|f(x_{n+1})| < \varepsilon_2$$

**Criterion 3 — Relative change is small:**

$$\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \varepsilon_3, \quad x_{n+1} \neq 0$$

In practice, Criterion 3 is preferred because it accounts for the scale of the answer. For example, an absolute change of 0.001 is small when  $x \approx 1000$ , but large when  $x \approx 0.001$ .

### 4.2 Order of Convergence

The **order of convergence**  $p$  of an iterative method describes how quickly the error decreases from one iteration to the next.

Let  $e_n = x_n - x^*$  be the error at step  $n$ . The method has **order**  $p$  if:

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^p} = C, \quad C > 0$$

where  $C$  is called the **asymptotic error constant**.

For large  $n$ , the definition means:

$$|e_{n+1}| \approx C|e_n|^p$$

The order  $p$  determines *how fast* convergence happens while the constant  $C$  affects speed.

- If  $p = 1$ : error shrinks by a constant factor (slow)
- If  $p > 1$ : faster than linear convergence (higher  $p$  usually means dramatically faster convergence. )

#### 4.2.1 How to Estimate $p$ and $C$ in Practice

In real problems, we do not know the exact solution  $x^*$ , but we can estimate:

1. Compute approximate errors:

$$e_n \approx |x_n - x_{n-1}|$$

2. Estimate the order:

$$p \approx \frac{\ln\left(\frac{e_{n+1}}{e_n}\right)}{\ln\left(\frac{e_n}{e_{n-1}}\right)}$$

3. Estimate the constant:

$$C \approx \frac{e_{n+1}}{e_n^p}$$

#### Example (Estimation)

Suppose:

$$e_1 = 0.1, \quad e_2 = 0.01, \quad e_3 = 0.0001, \quad \dots$$

Then:

$$e_2 = (e_1)^2, \quad e_3 = (e_2)^2, \dots$$

- So:

$$p = 2, \quad C = 1$$

This is **quadratic convergence**

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## 5 The Bisection Method

### 5.1 Motivation and Principle

The bisection method is the simplest and most robust root-finding method. It is based directly on the **Intermediate Value Theorem**: if  $f(a) \cdot f(b) < 0$ , a root exists in  $(a, b)$ . The method repeatedly **halves** the interval, discarding the half that does not contain the root. This is why the method is also called a **binary search method**.



## 5.2 Algorithm

**Given:**  $f$ , interval  $[a, b]$  with  $f(a) \cdot f(b) < 0$ , tolerance  $\varepsilon$

1. Compute midpoint:  $c = \frac{a+b}{2}$
2. If  $|b-a| < \varepsilon$  or  $|f(c)| < \varepsilon$ : **stop**,  $c$  is the root.
3. If  $f(a) \cdot f(c) < 0$ : set  $b = c$  (root is in left half  $[a, c]$ ). Else: set  $a = c$  (root is in right half  $[c, b]$ ).
4. Go to step 1.

## 5.3 Error Bound

After  $n$  bisection steps, the error satisfies:

$$|x_n - x^*| \leq \frac{b-a}{2^n}$$

- Accuracy improves **slowly but steadily** i.e the error decreases by a factor of **1/2 each step**.

- Convergence is **guaranteed** i.e Method is reliable but not efficient for high precision

**Number of iterations needed for tolerance  $\varepsilon$ :**

$$n \geq \frac{\ln\left(\frac{b-a}{\varepsilon}\right)}{\ln 2}$$

This tells us exactly how many iterations we need for a given accuracy.

## 5.4 Order of Convergence

The bisection method has **linear convergence**.

From:

$$|e_{n+1}| \approx \frac{1}{2}|e_n|$$

we identify:

- Order:  $p = 1$  (linear)
- Constant:  $C = \frac{1}{2}$

## 5.5 Solved Example

### Example 3.4

Find the root of  $f(x) = x^3 - x - 2 = 0$  in  $[1, 2]$  correct to 2 decimal places ,  $\varepsilon = 0.005$ .

**Step 0: Verify bracket.**  $f(1) = 1 - 1 - 2 = -2 < 0$  and  $f(2) = 8 - 2 - 2 = 4 > 0$ , therefore,  $f(1) \cdot f(2) < 0$

The interval width is  $b - a = 1$ . Number of steps we need:

$$n \geq \frac{\ln(1/0.005)}{\ln 2} = \frac{\ln 200}{\ln 2} \approx \frac{5.298}{0.693} \approx 7.6 \implies n = 8 \text{ iterations}$$

| Step | $a$    | $b$    | $c = \frac{a+b}{2}$ | $f(c)$       | New interval       |
|------|--------|--------|---------------------|--------------|--------------------|
| 1    | 1.0000 | 2.0000 | 1.5000              | $-0.125 < 0$ | $[1.5, 2.0]$       |
| 2    | 1.5000 | 2.0000 | 1.7500              | $1.609 > 0$  | $[1.5, 1.75]$      |
| 3    | 1.5000 | 1.7500 | 1.6250              | $0.666 > 0$  | $[1.5, 1.625]$     |
| 4    | 1.5000 | 1.6250 | 1.5625              | $0.252 > 0$  | $[1.5, 1.5625]$    |
| 5    | 1.5000 | 1.5625 | 1.5313              | $0.059 > 0$  | $[1.5, 1.5313]$    |
| 6    | 1.5000 | 1.5313 | 1.5156              | $-0.034 < 0$ | $[1.5156, 1.5313]$ |
| 7    | 1.5156 | 1.5313 | 1.5234              | $0.013 > 0$  | $[1.5156, 1.5234]$ |
| 8    | 1.5156 | 1.5234 | 1.5195              | $-0.011 < 0$ | $[1.5195, 1.5234]$ |

After 8 steps: root  $\approx 1.52$ , interval width  $= 0.0039 < 0.005$

**True root:**  $x^* = 1.5214$  (to 4 d.p.)

**Verification:**  $f(1.52) = (1.52)^3 - 1.52 - 2 = 3.512 - 3.52 = -0.008 \approx 0$

### Example 3.5

How many bisection steps are needed to find a root of  $f(x) = \cos(x) - x$  in  $[0, 1]$  accurate to 5 decimal places (i.e.,  $\varepsilon = 5 \times 10^{-6}$ )?

$$n \geq \frac{\ln\left(\frac{1-0}{5 \times 10^{-6}}\right)}{\ln 2} = \frac{\ln(200000)}{\ln 2} = \frac{12.206}{0.693} \approx 17.6$$

**Answer: 18 iterations.**

## 6 Tutorial Questions

### 6.1 Section A: Concepts and Definitions

#### Question 1

- Define the term *root* of a function  $f(x)$ . What does it mean geometrically?
- Explain why equations like  $e^x = 5x$  cannot be solved analytically, and must be handled numerically.

(c) Write each of the following in the form  $f(x) = 0$ , then determine whether a root exists in the given interval by evaluating  $f$  at the endpoints.

(i)  $e^x = x + 5$  on  $[1, 2]$

(ii)  $\ln(x) = 2 - x$  on  $[1, 2]$

(iii)  $x^3 = 5 - x$  on  $[1, 2]$

---

### Question 2

Explain the difference between the following pairs of terms:

(a) *Order of convergence* and *rate of convergence*

(b) *Convergence criterion* and *convergence condition*

(c) *Bisection method* and *Regula Falsi method*

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### Question 3

State the three stopping criteria commonly used in iterative root-finding methods. Explain when each one is appropriate and give an example scenario for each.

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## 6.2 Section B: Fixed-Point Iteration

### Question 4

The equation  $x^3 + x - 1 = 0$  can be rearranged in different ways:

(i)  $x = 1 - x^3$

(ii)  $x = (1 - x)^{1/3}$

(iii)  $x = \frac{1}{1 + x^2}$

For each rearrangement  $x = g(x)$ :

(a) Compute  $|g'(x)|$  at the approximate root  $x^* \approx 0.682$ .

(b) State whether the fixed-point iteration converges or diverges near  $x^*$ .

(c) For the convergent rearrangement(s), perform **5 iterations** starting from  $x_0 = 0.5$ . Record results in a table:  $n$ ,  $x_n$ ,  $x_{n+1}$ ,  $|x_{n+1} - x_n|$ .

---

### Question 5

The equation  $x = 2 \sin(x)$  has a root at  $x = 0$  (trivial) and non-trivial roots.

- (a) Rewrite as  $f(x) = 0$  and locate a non-trivial root in a suitable interval by sign-change test.
  - (b) Use fixed-point iteration with  $x_{n+1} = 2 \sin(x_n)$  from  $x_0 = 1.5$ . Perform 8 iterations. Does this converge?
  - (c) Check the convergence condition. Explain your observation.
- 

### 6.3 Section C: Bisection Method

#### Question 6

Use the bisection method to find a root of each function in the given interval. Carry out the iterations until the interval width is less than 0.01. Record each step in a table.

- (a)  $f(x) = x^3 - 2x - 5 = 0$  on  $[2, 3]$
  - (b)  $f(x) = x - e^{-x} = 0$  on  $[0, 1]$
  - (c)  $f(x) = \cos(x) - xe^x = 0$  on  $[0, 1]$
- 

#### Question 7

- (a) How many bisection iterations are needed to find a root of  $f(x) = x^2 - 7$  on  $[2, 3]$  to an accuracy of  $10^{-5}$ ?
  - (b) Carry out the required number of iterations and confirm your answer.
  - (c) Compare the result with  $\sqrt{7} = 2.6457513\dots$
- 

#### Question 8

The bisection method is applied to  $f(x) = x^5 - 3x - 1$  on  $[1, 2]$ .

- (a) Verify the sign change at the endpoints.
  - (b) After how many iterations is the root guaranteed accurate to 6 decimal places?
  - (c) Carry out **6 iterations** manually and find the approximate root.
- 

### 6.4 Section D: Secant and Regula Falsi Methods

#### Question 9

Apply the **secant method** to find the root of  $f(x) = x^2 - 5 = 0$ , starting from  $x_0 = 2$  and  $x_1 = 3$ .

- (a) Perform 5 iterations. Record  $n$ ,  $x_n$ ,  $f(x_n)$ ,  $x_{n+1}$ ,  $|x_{n+1} - x_n|$ .
  - (b) Compare your result with the true value  $\sqrt{5} = 2.2360679 \dots$
  - (c) Compute the ratio  $\frac{|e_{n+1}|}{|e_n|^{1.618}}$  for iterations 2 to 4. What does this suggest about the order of convergence?
- 

#### Question 10

Apply the **Regula Falsi method** to  $f(x) = e^x - 3x = 0$  on  $[1, 2]$ .

- (a) Perform 6 iterations. Record  $a$ ,  $b$ ,  $c$ ,  $f(c)$ , and the updated bracket at each step.
  - (b) Which endpoint stays fixed? What does this tell you about Regula Falsi?
  - (c) Compare the convergence speed with bisection applied to the same problem (compare how quickly  $|f(c)|$  decreases).
- 

### 6.5 Section E: Newton–Raphson Method

#### Question 11

Apply Newton–Raphson to find the root of each function, starting from the given  $x_0$ . Iterate until  $|x_{n+1} - x_n| < 10^{-6}$ .

- (a)  $f(x) = x^2 - 7$ ,  $x_0 = 3$
- (b)  $f(x) = \cos(x) - x$ ,  $x_0 = 0.5$
- (c)  $f(x) = x^3 - 3x + 1$ ,  $x_0 = 2$
- (d)  $f(x) = e^{-x} - x$ ,  $x_0 = 0.5$

For each: (i) Write down  $f'(x)$ . (ii) Show all iterations in a table. (iii) State how many iterations were needed.

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#### Question 12

The equation  $x - \tan(x) = 0$  has roots at  $x = 0$  and near  $x = 4.493$ .

- (a) Apply Newton–Raphson from  $x_0 = 4.5$  to find the non-trivial root.
  - (b) What happens if you start from  $x_0 = \pi/2$ ? Explain.
  - (c) What property of Newton–Raphson does this illustrate?
-

### Question 13

Consider  $f(x) = (x - 1)^2 = 0$ , which has a **double root** at  $x^* = 1$ .

- (a) Apply Newton–Raphson from  $x_0 = 2$ . Perform 8 iterations.
  - (b) Compute the errors  $e_n = |x_n - 1|$  and the ratio  $e_{n+1}/e_n$ .
  - (c) What order of convergence do you observe? Is this consistent with quadratic convergence?
  - (d) Explain theoretically why Newton–Raphson converges only **linearly** at a double root. *Hint:* Consider  $f'(x^*)$  for a double root.
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## 6.6 Section F: Newton’s Method for Polynomials and Horner’s Scheme

### Question 14

Use Horner’s scheme to evaluate  $P(x) = 3x^4 - 2x^3 + x^2 - 5x + 4$  and  $P'(x)$  at:

- (a)  $x = 2$
- (b)  $x = -1$
- (c)  $x = 0.5$

For each, show the full Horner table (both passes) and verify by direct calculation.

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### Question 15

Use Newton’s method (with Horner’s scheme) to find a root of  $P(x) = x^4 - 3x^3 + x^2 + 5x - 2$  near  $x_0 = 3$ .

Perform **4 iterations** and record  $x_n$ ,  $P(x_n)$ ,  $P'(x_n)$ ,  $x_{n+1}$  at each step.

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## 6.7 Section G: Comparison and Reflection

### Question 16

The equation  $f(x) = x^3 - 7x + 2 = 0$  has three real roots.

- (a) Show by sign changes that roots exist in  $[-3, -2]$ ,  $[0, 1]$ , and  $[2, 3]$ .
- (b) Apply bisection (5 iterations each) to find each root to 2 decimal places.
- (c) Apply Newton–Raphson to find each root, choosing appropriate starting values. Compare the number of iterations needed with bisection.

- (d) For the root in  $[0, 1]$ , compare the convergence of all four methods (bisection, Regula Falsi, secant, Newton–Raphson) by applying each and tabulating the errors at each iteration.
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**Question 17**

Apply Bairstow's method to find all roots of  $P(x) = x^3 - 6x^2 + 11x - 6$ .

- (a) Start with  $r = 1$ ,  $s = -2$ . Perform two full Bairstow iterations (computing  $b_k$ ,  $c_k$ ,  $\Delta r$ ,  $\Delta s$  at each step).
- (b) Once a quadratic factor  $x^2 - rx - s$  is found, extract the two roots from it.
- (c) Divide  $P(x)$  by the quadratic factor to find the remaining linear factor and its root.
- (d) Verify all three roots by substitution into  $P(x)$ .

(Note: The roots of this polynomial are  $x = 1, 2, 3$  — use this to check your work.)

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*End of Topic 3 Tutorial Questions*

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